SO(3) Equivariant Neural Networks

Lecture 5

Build a Physical SO(3) ENN

Training, validation, and the path to many-particle systems

90-minute journal-club lecture note

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Companion notebook included in the series package

Session map and learning goals



By the end of this lecture, you should be able to:

- assign irrep types to a physical model's inputs and outputs;
- design a dataset that separates invariant shape from global orientation;
- train and independently test an equivariant model;
- explain why equivariance improves rotational generalization but does not guarantee accuracy;
- sketch an SO(3)-equivariant message-passing model for a many-particle system.

Key idea

Building a physical ENN begins with transformation laws, not code: identify the geometric type of each quantity, choose allowed intermediate irreps, couple them with intertwiners, and test the claimed symmetry independently of the loss.

1 A dipole-inspired physical task

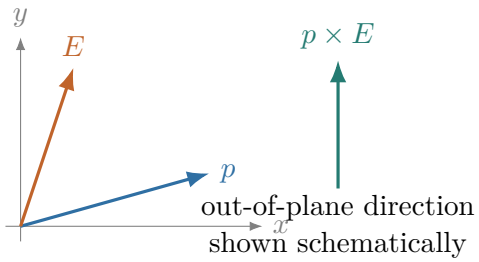
Let p be a dipole moment and E an external field. Both transform as $\ell = 1$ vectors. Familiar outputs include the scalar interaction energy and vector torque:

$$U(p, E) = -p \cdot E, \qquad \tau(p, E) = p \times E. \tag{1}$$

The notebook uses the nonlinear torque-like response

$$y(p, E) = (p \cdot E)(p \times E). \tag{2}$$

This requires both the scalar and vector pieces of $1 \otimes 1$ and produces an $\ell = 1$ output.



Physics lens

The target is a pedagogical constitutive response rather than a universal dipole law. Its purpose is to require nonlinear interaction between an invariant alignment measure and an equivariant direction. The architecture, not the particular target, is the transferable lesson.

2 Design the feature types before the network

The design worksheet is:

Role	Quantity	Type
Input	dipole p	$\ell = 1$
Input	field E	$\ell = 1$
Hidden	alignment and other invariants	$\ell = 0$
Hidden	oriented responses	$\ell = 1$
Hidden	anisotropic alignment	$\ell = 2$
Output	nonlinear torque-like response y	$\ell = 1$

The notebook chooses

$$\{1:2\} \longrightarrow \{0:2, 1:2, 2:2\} \longrightarrow \{1:1\}. \tag{3}$$

This is small enough to inspect completely while still containing the required interaction paths.

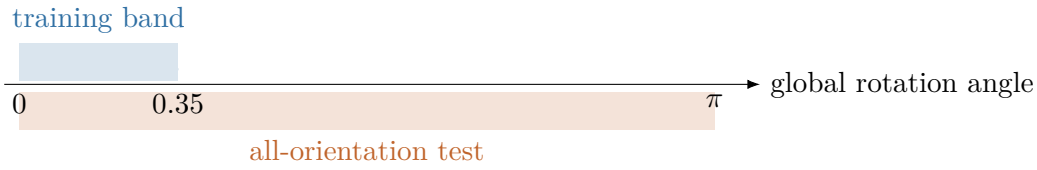
3 Dataset design: shape versus orientation

For two unit vectors, their invariant shape is determined by the relative angle, or equivalently

$$\cos\theta = p \cdot E. \tag{4}$$

A global rotation changes orientation without changing this relative angle.

The training set samples every relative angle but applies only small global rotations, up to 0.35 radians. The test set covers global rotations up to π.



This experiment asks whether the model can generalize across a transformation known to be physically irrelevant to the underlying relationship.

AI lens

An equivariant model cannot assign different laws to two globally rotated copies of the same physical configuration. A plain MLP can, unless data and optimization happen to teach it otherwise.

4 Training and evaluation

The loss is an ordinary Cartesian mean squared error:

$$\mathcal{L}(\theta) = \frac{1}{3N} \sum_{n=1}^{N} \left\| f_\theta(p_n, E_n) - y_n \right\|^2. \tag{5}$$

The notebook uses finite-difference gradients to keep the implementation in NumPy. In a practical model, automatic differentiation would replace this plumbing without changing the equivariant layer.

Prediction error and equivariance error answer different questions:

$$\text{MSE}: \quad \text{Does the model approximate the target?} \tag{6}$$

$$\epsilon_{\text{eq}}: \quad \frac{\max |f(Rp, RE) - Rf(p, E)|}{\max |Rf(p, E)|}. \tag{7}$$

Model state	MSE	Equivariance error
Accurate and equivariant	small	small
Inaccurate but equivariant	large	small
Accurate only on training orientations	small on train	large
Completely poor model	large	large or small

The notebook pins its data seed, so the numbers below are reproducible. The baselines are averaged over five initializations, because a single run of the unconstrained model is not a result:

Model	Train MSE	All-orientation MSE	Eq. error
Spherical CG network	4×10^{-5}	4×10^{-5}	6×10^{-16}
Plain Cartesian MLP	$3.3(2) \times 10^{-3}$	$1.6(8) \times 10^{-1}$	~ 1
MLP $+8\times$ rotations	$1.2(1) \times 10^{-2}$	$4.1(2) \times 10^{-2}$	~ 1

Read the spread on the plain MLP first: $\pm 50\%$ across seeds on the very number the comparison rests on. Any single run of that row would have been noise reported as a finding.

4.1 Would augmentation have done the same job?

The third row answers the question Lecture 1 raised and could not yet measure. Training the baseline on eight globally rotated copies of every pair recovers most of the orientation generalization: the all-orientation error drops by a factor of four, at eight times the data and eight times the fitting cost, and at the price of a worse fit on the original examples.

What augmentation does *not* do is move the equivariance residual. It stays at order unity. The augmented model has learned to behave approximately the same way on rotated inputs it has seen; it has not acquired the transformation law, and nothing prevents it from violating that law between training examples. The equivariant network's residual is 10^{-16} for every input and every rotation, including ones no one will ever test.

Key idea

Augmentation and architecture are not two routes to the same guarantee. Augmentation buys accuracy on the orbit you sampled. Architecture buys an identity that holds everywhere, and it happens to buy accuracy too.

4.2 Which paths does the trained model use?

The architecture carries scalar, vector and quadrupole channels, but whether it *needs* them is an empirical question. Setting one learned path matrix to zero at a time and re-measuring gives:

Path removed from the output layer	Train MSE	Test MSE
nothing (full model)	4×10^{-5}	4×10^{-5}
$0 \otimes 1 \to 1$	4.8×10^{-2}	5.3×10^{-2}
$1 \otimes 1 \to 1$	4×10^{-5}	4×10^{-5}
$1 \otimes 2 \to 1$	3.4×10^{-4}	3.8×10^{-4}
$2 \otimes 2 \to 1$	5.5×10^{-4}	6.5×10^{-4}
same-ℓ linear path	2.8×10^{-2}	2.8×10^{-2}
every path producing $\ell = 2$	1.6×10^{-3}	1.9×10^{-3}

Two lessons, and the second one is the more useful. First, $\ell = 2$ is not decoration: deleting the quadrupole channel at its source costs well over an order of magnitude, so those five components carry real information about the target. Second, single-path ablation measures *replaceability*, not importance. Removing $1 \otimes 1 \to 1$ alone costs nothing measurable, because the linear path and the remaining couplings can absorb its job; that is a statement about redundancy in the architecture, not about the coupling being useless. Ablation studies in the literature are read wrongly for exactly this reason.

Note also that train and test degrade together in every row. An ablated equivariant network is still exactly equivariant. Removing capacity costs accuracy; it cannot cost symmetry.

4.3 What equivariance costs

The equivariant model fits 60 parameters in roughly 20 seconds here, against 643 parameters in under a second for the baseline. Most of that gap is the finite-difference plumbing rather than the architecture, and automatic differentiation would erase it. The real cost appears at scale: the CG tensor products dominate the compute of a production model, and the number of allowed paths grows sharply with $\ell_{\max}$. Buying an exact symmetry is not free, and it is worth knowing what it is being bought with.

5 A disciplined validation protocol

For any physical ENN, test at several levels:

1. **Representation checks:** $D(R_1)D(R_2) = D(R_1R_2)$ and unitarity.
2. **Coupling checks:** each CG tensor satisfies the intertwining identity.
3. **Layer checks:** randomized inputs and weights commute with transformations.
4. **End-to-end checks:** test the complete untrained and trained network.
5. **Physical checks:** units, limiting cases, conservation laws, and known analytic results.
6. **Generalization checks:** separate unseen orientations from unseen physical shapes.

Pause and discuss

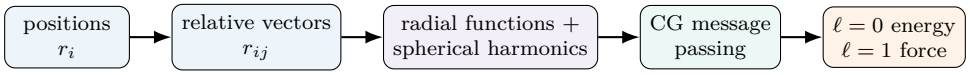
Why should equivariance be tested before training? If the error is already large for random weights, what kinds of implementation bugs would you suspect?

6 From two vectors to a many-particle physical system

For particles at positions r_i, use relative displacements

$$r_{ij} = r_j - r_i \tag{8}$$

to remove global translation. A common equivariant message-passing pipeline is



The symmetry responsibilities are divided as follows:

- relative vectors give translation invariance;
- radial functions depend only on $\|r_{ij}\|$;
- spherical harmonics $Y_{\ell m}(\hat{r}_{ij})$ encode direction as irreps;
- CG products combine angular feature types;
- neighbor sums give permutation invariance to ordering;
- an $\ell = 0$ readout predicts energy, while $\ell = 1$ features can predict forces directly.

If forces are defined as $F_i = -\nabla_{r_i} E$, rotational invariance of the scalar energy implies equivariant forces, provided differentiation is performed consistently.

7 Connection to neural quantum states

A neural quantum state represents a wavefunction $\psi_\theta(x)$ and optimizes the variational energy

$$E_\theta = \frac{\langle \psi_\theta | H | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle}. \tag{9}$$

Rotational symmetry can be handled softly, for example with a Casimir penalty, or structurally by making the wavefunction transform in a chosen total-angular-momentum irrep. A rotational ground state often lies in an $\ell = 0$ sector, while excited multiplets may carry $\ell > 0$ and magnetic components m. The same representation and CG ideas then organize how particle features combine inside the NQS.

Physics lens

One loose end from Lecture 1 is worth closing here. The recurring target $(p \cdot E)(p \times E)$ contains a cross product, so it is a *pseudovector*: under an improper transformation it acquires an extra sign relative to an ordinary vector. Throughout the series we test proper rotations only, where polar and axial vectors are indistinguishable, so nothing has gone wrong. But a model of a real system that must respect reflections needs O(3) rather than SO(3), which means carrying a parity label

alongside ℓ and letting it decide which couplings are allowed. Forces are polar; torques and magnetic moments are axial. Choosing the wrong one is a symmetry bug that no rotation test will find.

Math lens

Equivariance does not solve the variational problem. It restricts the trial-state family to the desired symmetry behavior. Expressivity, antisymmetry for fermions, sampling quality, and optimization remain separate challenges.

8 Capstone design activity

Choose one system: molecular forces, rigid-body torque, magnetic response, material stress, or fluid velocity. In a small group, answer:

1. What are the inputs and their transformation types?
2. What output is required, and what is its ℓ type?
3. Which symmetries besides rotation matter?
4. Which intermediate ℓ values are likely useful?
5. Which CG paths can create the desired output?
6. Where can nonlinearities enter safely?
7. What baseline and data split would reveal the value of equivariance?
8. What numerical test would falsify the equivariance claim?

9 Questions and discussion

Question 5.1. Why does the restricted-orientation dataset cover physical shape more fully than its visual appearance might suggest?

Question 5.2. Why should prediction MSE and equivariance error be reported separately?

Question 5.3. For the dipole problem, assign irrep types to p, E, U, and τ. Which $1 \otimes 1$ paths produce U and τ?

Question 5.4. How do relative displacements, neighbor aggregation, spherical harmonics, and CG products address different symmetries in a particle model?

Question 5.5. If an invariant energy model is differentiated to obtain forces, why should the forces be equivariant? State any assumptions.

Question 5.6. In a neural quantum state, what is the difference between encouraging an angular momentum sector with a penalty and building the sector into the architecture?

Question 5.7. The augmented baseline improves its all-orientation error by a factor of four while its equivariance residual does not move. Explain both halves of that result.

Question 5.8. Removing the $1 \otimes 1 \to 1$ path from the trained model changes its error by nothing measurable. Give two different explanations consistent with this observation, and describe an experiment that would distinguish them.

Checkpoint

Series goal: you should now be able to sketch an SO(3) ENN by typing the physical quantities, selecting irreps and CG paths, adding symmetry-safe nonlinearities, and planning independent equivariance tests.